

Upper Crossed Syndrome and Postural Considerations



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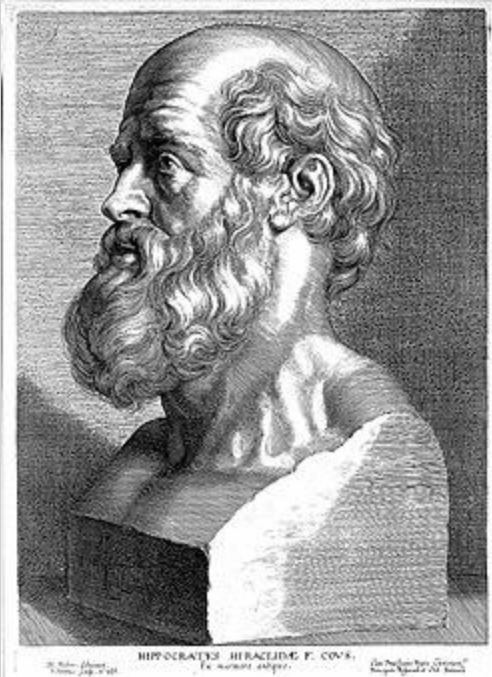
Objectives

1. the students will be able to appreciate the historical background/perspective, from Hippocrates, to A.T. Still, to consideration of the 5 (or 6) Osteopathic Models, to Dr. Phil Greenman's legacy, and on to the related to inter-professional and global awareness of the Upper (and Lower) Crossed syndrome, from the legacy of neurologists, Drs. Janda and Lewit.
2. the students will be able to understand the Upper Crossed Syndrome, including the role of the type I (slow twitch, postural) and type II (fast twitch, phasic) muscles. The student will understand which muscles need stretching, and which muscles need strengthening, in relation to application of an understanding of this "crossed" syndrome. Specific exercises will be reviewed at the end of the lecture, with pictures and diagrams included.
3. the students will be able to understand the importance of obtaining and considering the patient's history (subjective), along with objective information. This objective information can include Special Tests, muscle testing/palpation, the osteopathic structural examination, postural observation and analysis, and other neurological (NCV / EMG) or radiological (x-rays, MRI, etc.) testing, as needed.
4. the students will be able to understand how to incorporate an understanding of postural compensation, tensegrity (a.k.a. biotensegrity), gravitational strain pathophysiology (G.S.P.) and postural decompensation into their osteopathic perspective.
5. the students will read the assigned article from Page (LINK: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3201065/>), which is also listed under the Resources for this Classroom Session) on their own, to gain a sense of the clinical correlation / clinical suggestion related to this topic area.

Outline

- Historical perspective
- Review Muscle Types (I and II)
- Upper Crossed Syndrome
- Postural Considerations
 - Postural Compensation
 - Tensegrity
 - Gravitational Strain Pathophysiology (G.S.P.) and Postural Decompensation
- General Treatment Strategies

Historical Perspective



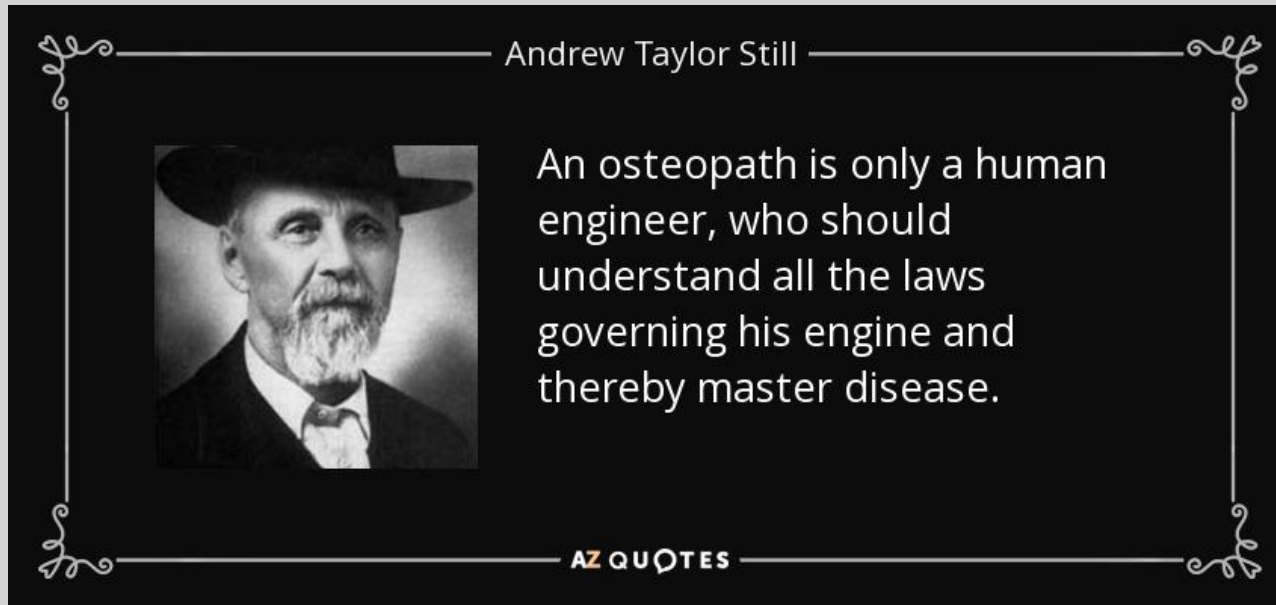
Engraving: bust of Hippocrates by [Paulus Pontius](#) after [Peter Paul Rubens](#), 1638
<https://share.google/3NOJhdAazy25CJg9l>

“Nearly 2,500 years ago, **Hippocrates** said, ‘**The regimen I adopt will be for the benefit of my patients according to my abilities and judgment.**’”

As we learn about Postural Considerations in Osteopathic Diagnosis and Treatment, we can consider:

- “principles intended to enhance the ability to understand postural mechanisms as dynamic processes affecting interrelated patient neuromusculoskeletal function.”
- that “with understanding and practice, the physician’s judgment in prescribing individualized postural treatment regimens will evolve and improve.”

Historical Perspective

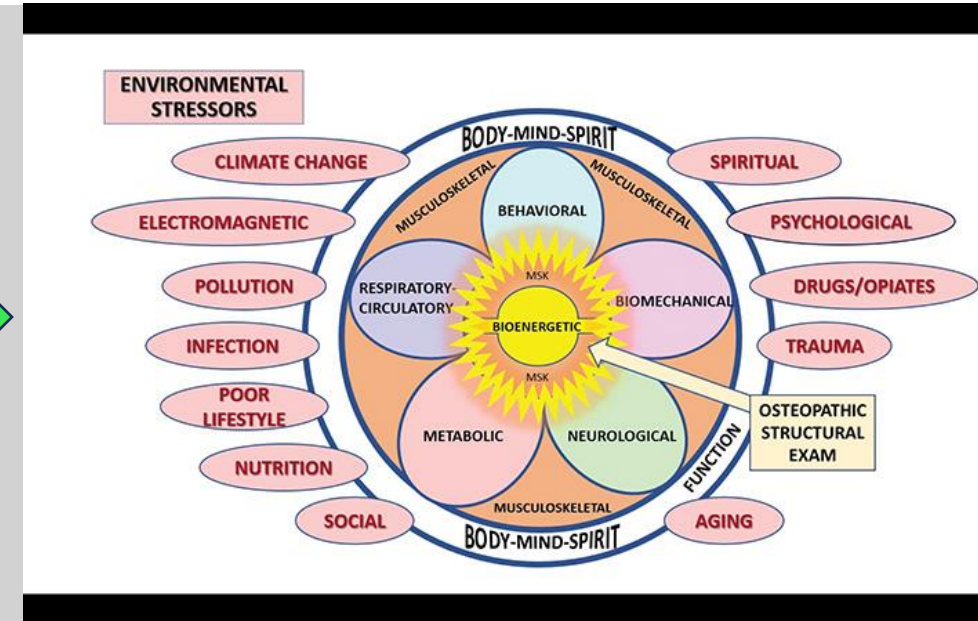
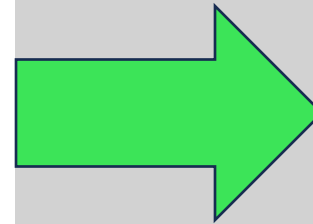
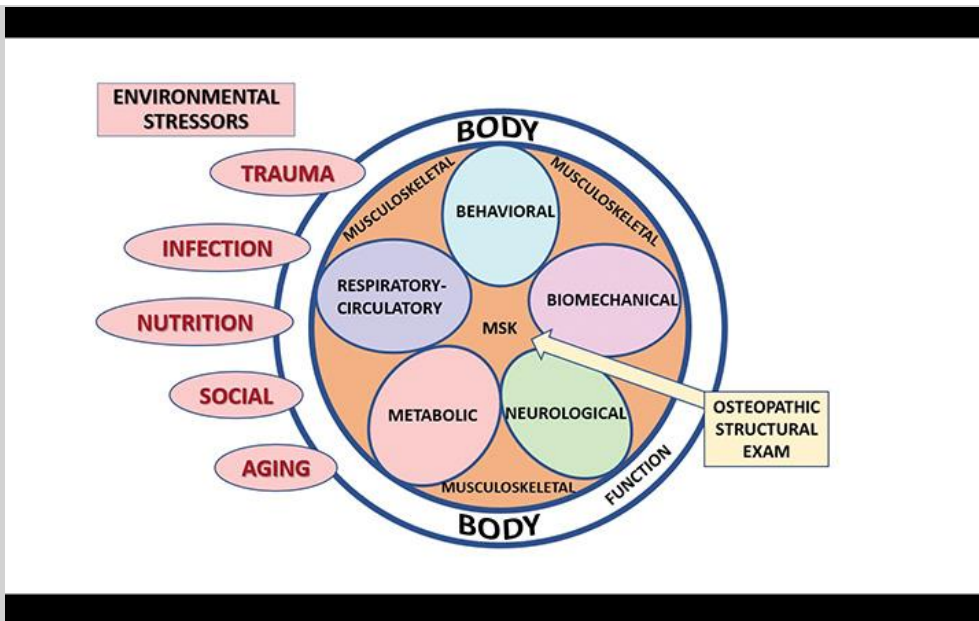


<https://share.google/2j2scczUSyT2AvyUs>
<https://www.lifemedidaho.com/osteopathy>

“Andrew Taylor Still placed much emphasis on **the fascia and its relationship to health**. He wrote, ‘I know of no other part of the body that equals the fascia as a hunting ground (sic. for health/disease) ... By its action we live, and by its failure, we die.’”

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model**.
Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 5e, 2026,
LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456&resultClick=1#261507713>

Historical Perspective



Original five (5) models adopted
by Educational Council on Osteopathic
Principles (ECOP) in 1982.

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model.**
Foundations of Osteopathic Medicine: Philosophy,
Science, Clinical Applications, and Research, 5e, 2026

Department of Osteopathic Manipulative Medicine

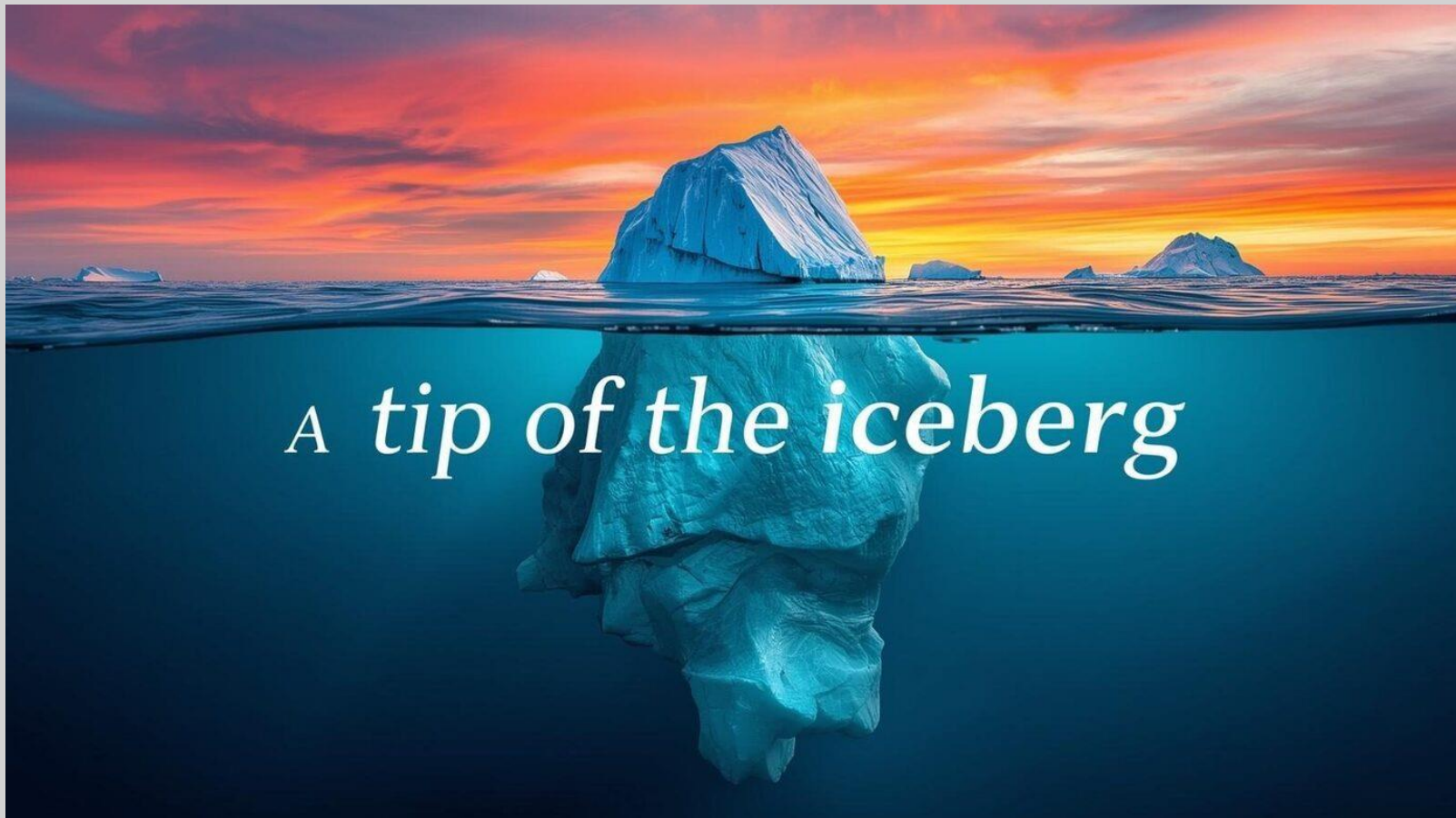
**Expanded six (6) models to
include bioenergetics.**

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model.**
Foundations of Osteopathic Medicine: Philosophy,
Science, Clinical Applications, and Research, 5e, 2026,
LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456&resultClick=1#261507624>

Historical Perspective

NEW YORK INSTITUTE
OF TECHNOLOGY

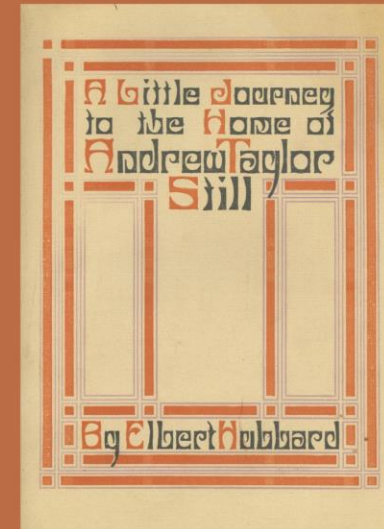
College of Osteopathic
Medicine



Historical Perspective

D.O. = “Dig On”
(i.e., continue to
search
and learn!)
in your osteopathic
reasoning!

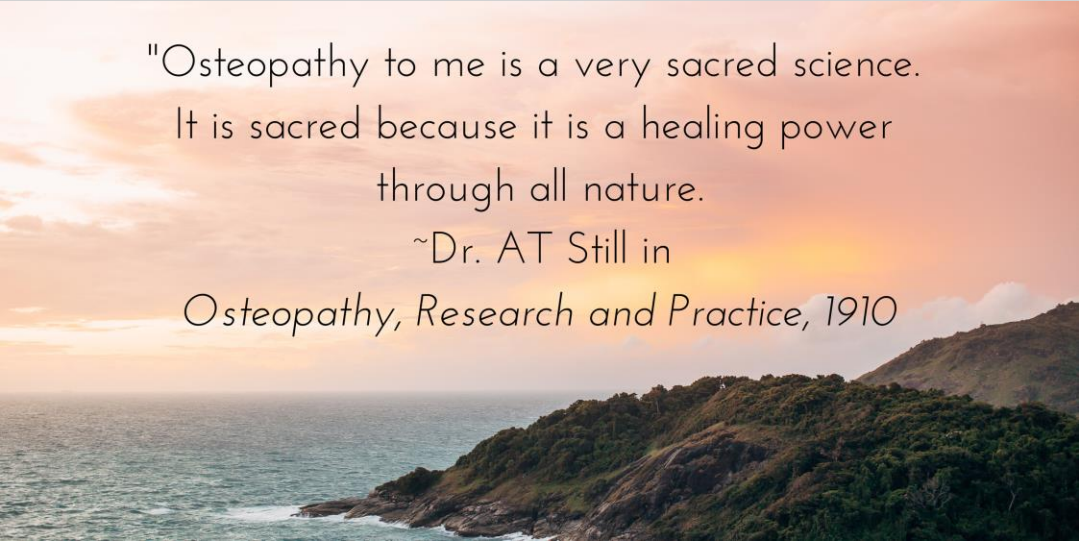
In 1909, the famous publisher and author Elbert Hubbard visited “The Old Doctor” and published this 32-page book which helped to spread the word of Osteopathy.



“Osteopathy is simply the practice of common sense.” ~Elbert Hubbard

Historical Perspective

D.O. = “Dig On” (i.e., continue to search and learn!) in your osteopathic reasoning!



"Osteopathy to me is a very sacred science.
It is sacred because it is a healing power
through all nature.
~Dr. AT Still in
Osteopathy, Research and Practice, 1910

<https://share.google/BzQ8yZdbEBnjVtBTp>

➤ Your “digging on” (your search) may bring you to incorporate information and/or practical approaches, which originate from other fields of reasoning, including but not limited to osteopathic medicine itself; neurology, orthopedics, physiatry, functional medicine, or any other medical specialty or primary care field; bioenergetics, spirituality, metabolism/nutrition, or any of the other components of the “**6 Models**” of Osteopathic Medicine; chemistry, physics (quantum physics), or any of the other “basic” (pre-med) or advanced level sciences; embryology, growth and development, etc.

Historical Perspective

NEW YORK INSTITUTE
OF TECHNOLOGY

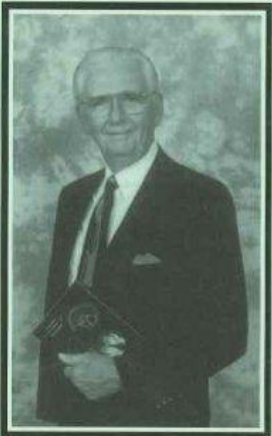
College of Osteopathic
Medicine

Philip E. Greenman, DO, FAAO
(1928-2013)

→ “Digging On” in Osteopathic
Medicine throughout his career

GREENMAN'S WORKS

THE COLLECTED WORKS OF
PHILIP E. GREENMAN, DO, FAAO



“What I do know is that when I put my hands on a patient as part of a comprehensive physical examination, I begin to evolve a line of questioning that isn’t entirely conscious. I pick up information in my hands that guide the questions I next need to ask.”

Philip E. Greenman, D.O., FAAO
Michigan State University COM

Dr. Greenman was also aware of the work of Drs. Janda and Lewit and the Crossed Syndromes.

Historical Perspective

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Philip E. Greenman, DO, FAAO
(1928-2013)



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Department of Osteopathic Manipulative Medicine

Philip E. Greenman, DO, FAAO



Dr. Greenman taught osteopathic manipulative medicine at all levels - osteopathic students, interns, residents, fellows, allopathic and osteopathic physicians, and other health care professionals - and achieved international recognition for his teaching in manual medicine tutorials.

His video series is the most complete and comprehensive set of educational materials that is available to practitioners of manual medicine. In these videos, Dr. Greenman introduces the student to basic models of manual medicine as they related to diagnosis and treatment, guided the student through a standard screening examination of a patient, introduced the student to the unique anatomic/biomechanic characteristics of specific regions within the musculoskeletal system, and finally guided the student through the application of comprehensive diagnostic and treatment protocols.

Now deceased, Dr. Greenman served as senior associate dean and professor of osteopathic manipulative medicine and of physical medicine and rehabilitation at the Michigan State University College of Osteopathic Medicine. A charter faculty member at MSUCOM, Dr. Greenman is recognized for both his artistry and scientific knowledge of manual medicine.

"I'd decided that I wanted to be a DO at age eight," Greenman remembered. "Our family doc was an '03 graduate of the Kirksville college, then called the American School of Osteopathy. As a second grader, I saw one of my classmates experience back and leg pain and weakness after a fall. They took her to Albany for treatment, but she didn't get better until she was seen by a DO." From that early decision

<https://share.google/iRSLFYEvqxRhRD8UK>

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)

- H/o poliomyelitis as a teenager, wheelchair-bound for 3 years after an initial quadriplegia.
- **Neurologist** training at Charles University, Prague.
- 1960s in Canada, he did post-doctoral work with Professor John Basmajian.
- His work in rehabilitation included research and clinical activities, over many decades and with a global influence.
- He was the Editor in Chief of the Czech Journal, Rehabilitation and Physical Medicine, until 2002.
- He worked with **Karel Lewit, M.D.**, another Czech **neurologist**, **presenting the Upper & Lower Cross Syndromes.**



Vladimir Janda, MD, DSc 1928–2002.

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)



Vladimir Janda, MD, DSc 1928–2002.

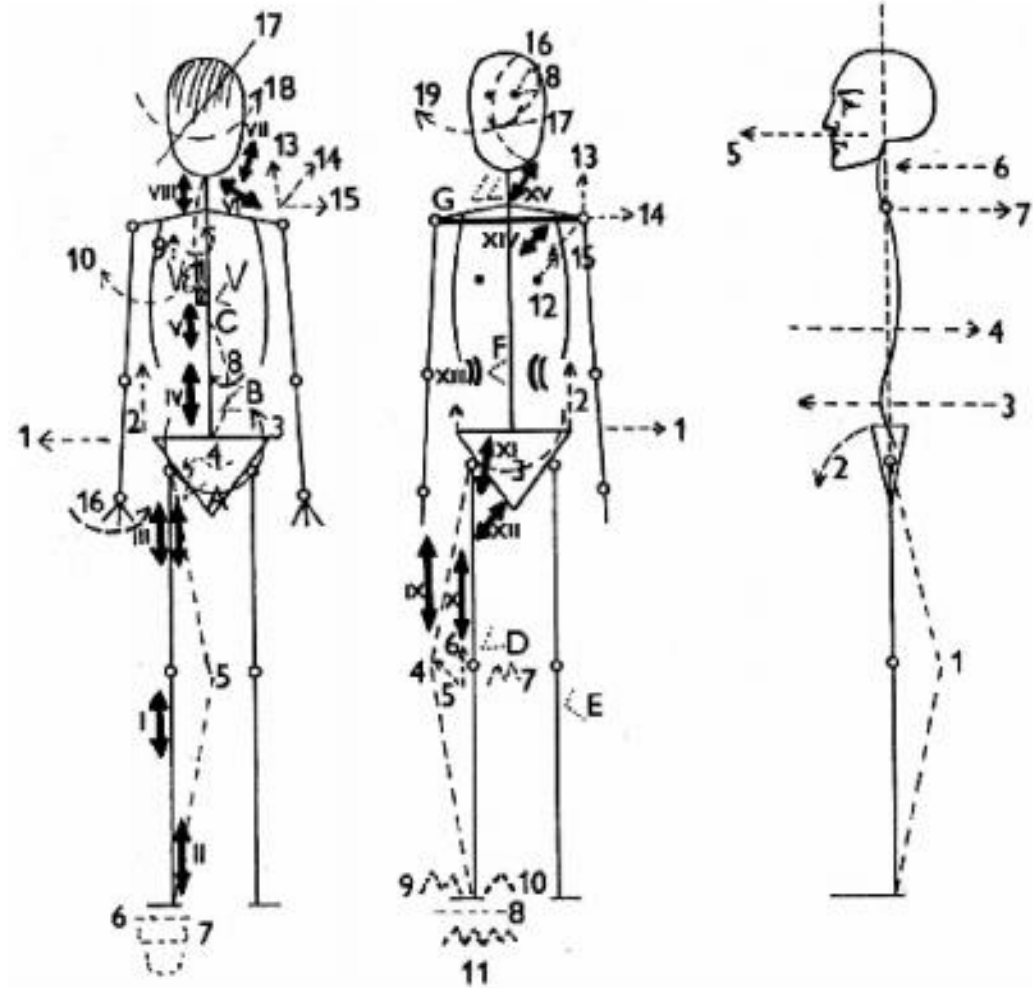
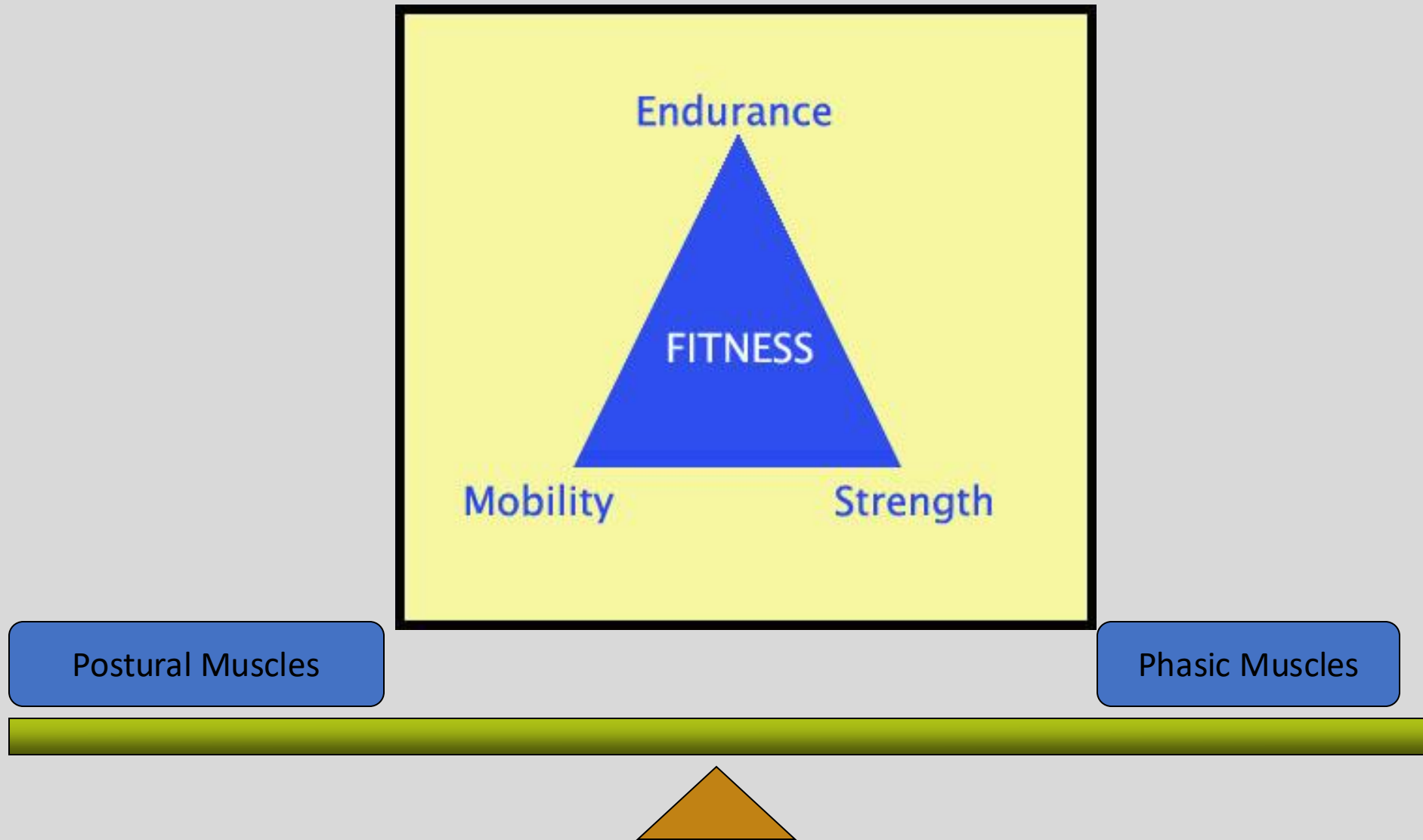


Figure 7. One of Janda's drawings demonstrating postural assessment. He emphasized the local and global assessment as part of a complete examination, teaching methods that allowed the examiner to quickly log his or her findings. (Reproduced from Janda V. Dokumentace analyzy stoje [How to record stance analysis]. *Rehabilitace a Fyzikalni lekarstvi* 1994:4–5, with permission.)

Balance



Which muscles are used?



Ashlyn Wang

The leading runners of the first wave run down Beacon Street in Brookline, Mass., toward the finish line on Monday, April 15, 2024. (Ashlyn Wang/Beacon Staff)

<https://share.google/Oz2tlxzlkbxH5qhX>

- **Type I – tonic, slow-twitch muscles – “long endurance work” – postural muscles – tend to become facilitated and shortened/tight**



Picture by Steph Chambers/WA/Getty Images for World Athletics 2026

<https://www.olympics.com/en/news/tate-taylor-world-u20-champion-one-upping-noah-lyles>

- **Type II – phasic, fast-twitch muscles – fatigue more easily – tend to become inhibited and muscles become weak**

In clinical practice, we distinguish two muscle types in the evaluation and treatment:

**Type I muscles = postural muscles =
slow twitch muscles**

These muscles react by becoming TIGHT or SHORTENED as they become “facilitated” in response to postural imbalance, injury, somatic dysfunction.

Thus it is good to know which muscles are being categorized as postural/slow twitch muscles.

**Type II muscles = phasic muscles =
fast twitch muscles**

These muscles react by becoming WEAK as they become “inhibited,” For instance, the Gluteus medius muscle, a phasic muscle, may become reflexly inhibited due to an associated sacroiliac joint issue (e.g. extended, torsion, etc.) on one side and thus causing the muscle to become relatively weak on that side vs. the other side. The muscle itself has “not lost its innate strength” but rather is inhibited and thus becomes weak as a result.

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
TYPE I	Postural	Slow twitch	Become facilitated and thus SHORTENED or TIGHT	Upper crossed syndrome: Example 1: Head forward posture – <u>Anterior tight/shortened</u>= facilitated muscles: Pectorales <u>Posterior tight /shortened</u>=facilitated muscles: Levator scapulae, upper trapezius, SCM Example 2: when there is excessive ankle pronation – one can see the lower crossed syndrome – <u>Anterior: tight</u> iliopsoas muscle and <u>Posterior: tight</u> erector spinae muscles

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
Type II	Phasic	Fast twitch	Become <u>inhibited</u> and thus WEAK	Upper crossed syndrome (example: head forward posture) <u>Anterior: weak</u> neck flexors=inhibited <u>Posterior: weak</u> rhomboids/serratus anterior LOWER crossed syndrome: <u>Anterior: weak/inhibited</u> abdominals <u>Posterior: weak/inhibited</u> gluteus maximus/medius

Upper Crossed and Lower Crossed Syndromes

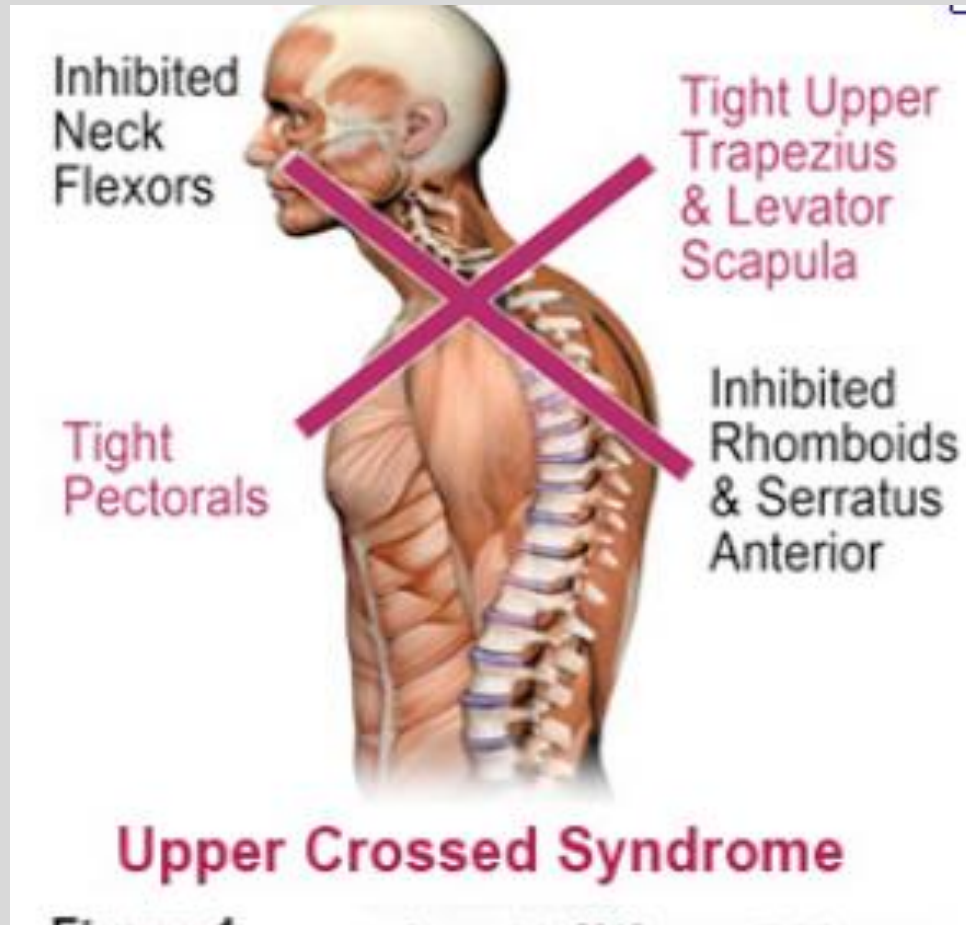
Clinically (e.g. work by Vladimir Janda, M.D./Prague/Czech Republic) there have been observed PATTERNS of muscle involvement, that have been called the **upper and lower crossed syndromes**. “Upper” because it involves the upper body, and “lower” because it involves the pelvis, or lower body.

It is called “Crossed” as there are two ‘lines’ that form a cross-over with ANTERIOR muscles being facilitated=tight=shortened (type I muscles) and POSTERIORLY (follow one LINE – call it the “facilitated” or “shortened” line) with type I muscles being also shortened. This is in response to a muscle imbalance.

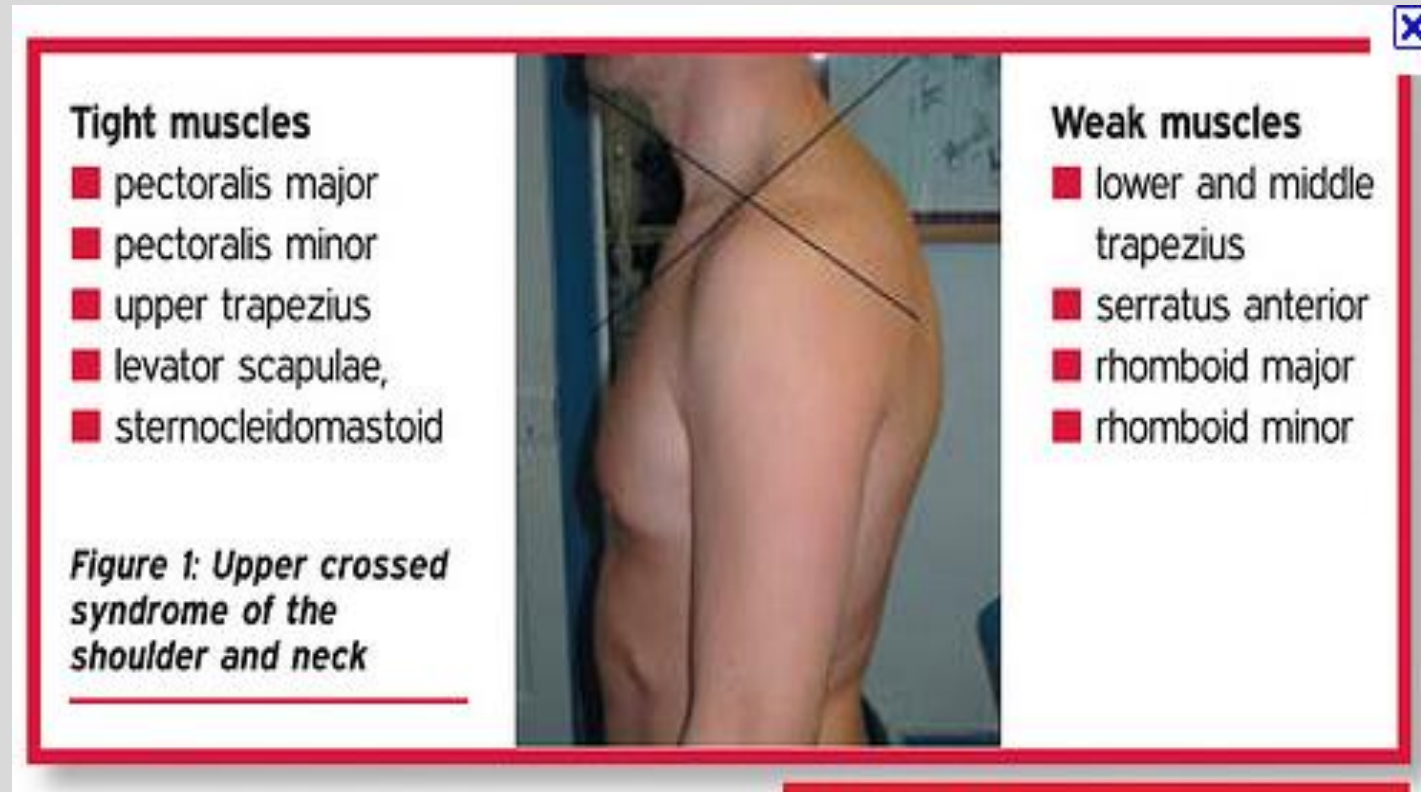
Similarly, follow the other cross-line that goes from ANTERIOR to POSTERIOR, representing the INHIBITED and thus WEAK muscles both anteriorly and posteriorly.

So it is like a “puzzle” where you have two lines with four “stations:” 2 anterior and 2 posterior groups. Knowing the four “groups” – will give you an understanding of which muscles can be involved at any one time – so if the pectoralis major is tight in front, and you do have a cross syndrome – upper, for instance – you can deduce which other muscles are involved in a particular patient.

Upper Crossed Syndrome - Posture and muscle imbalance



Upper crossed syndrome



POSTURE and The **CROSSED SYNDROMES**

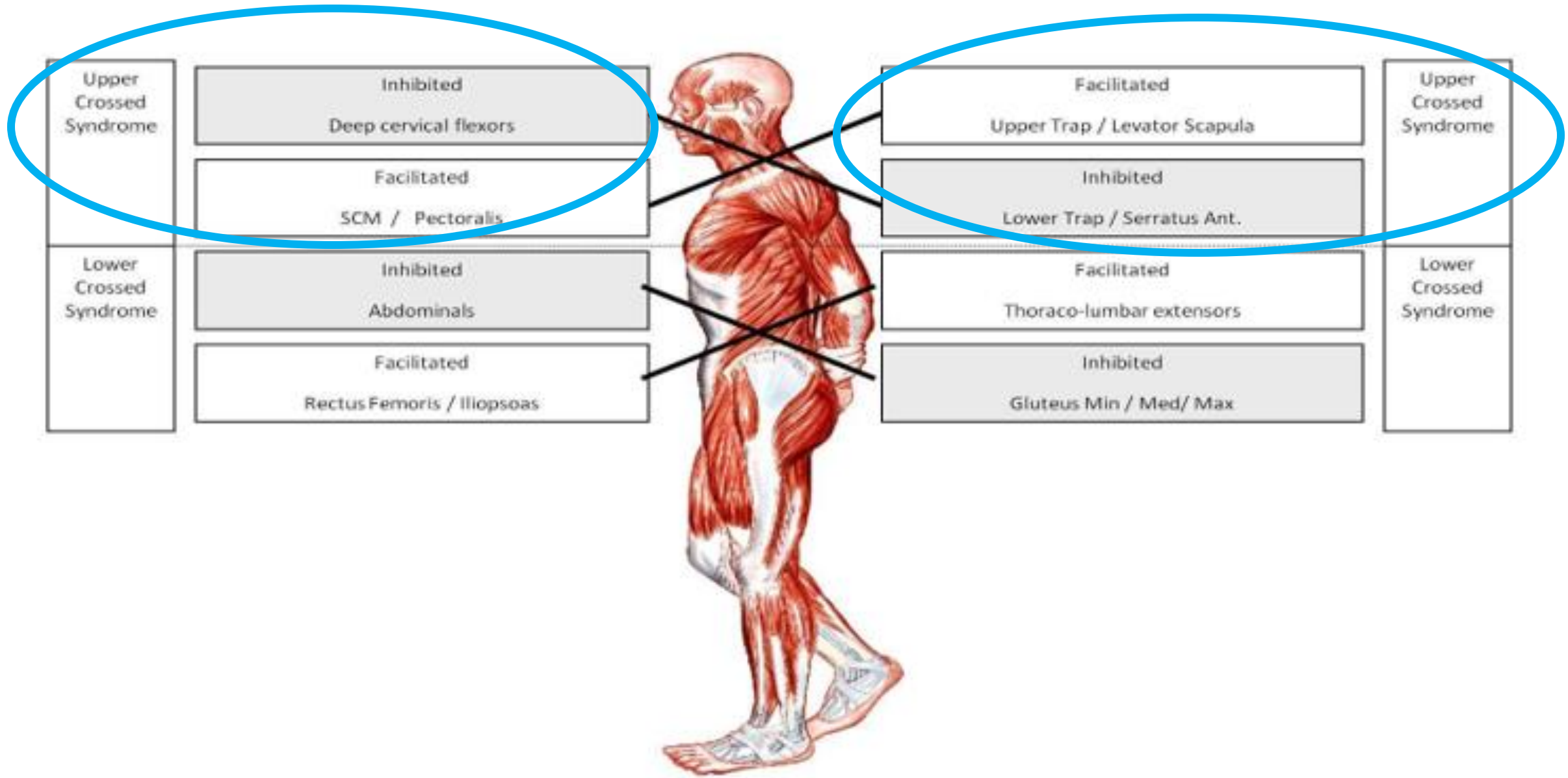


Figure 1 : Janda's Muscle Imbalance Syndromes

Table 1 – Anterior Muscles involved in Upper (and Lower) Crossed Syndromes

memorize

Anterior Postural Muscles	Anterior Phasic Muscles
Upper Cross	
Scalenes	Deep cervical flexors
Sternocleidomastoid	
Pectoralis major (upper)	
Pectoralis minor	
Lower Cross	
Iliopsoas	Rectus abdomini
Rectus femorus	Three Vasti muscles
Adductor brevis	Tibialis anterior

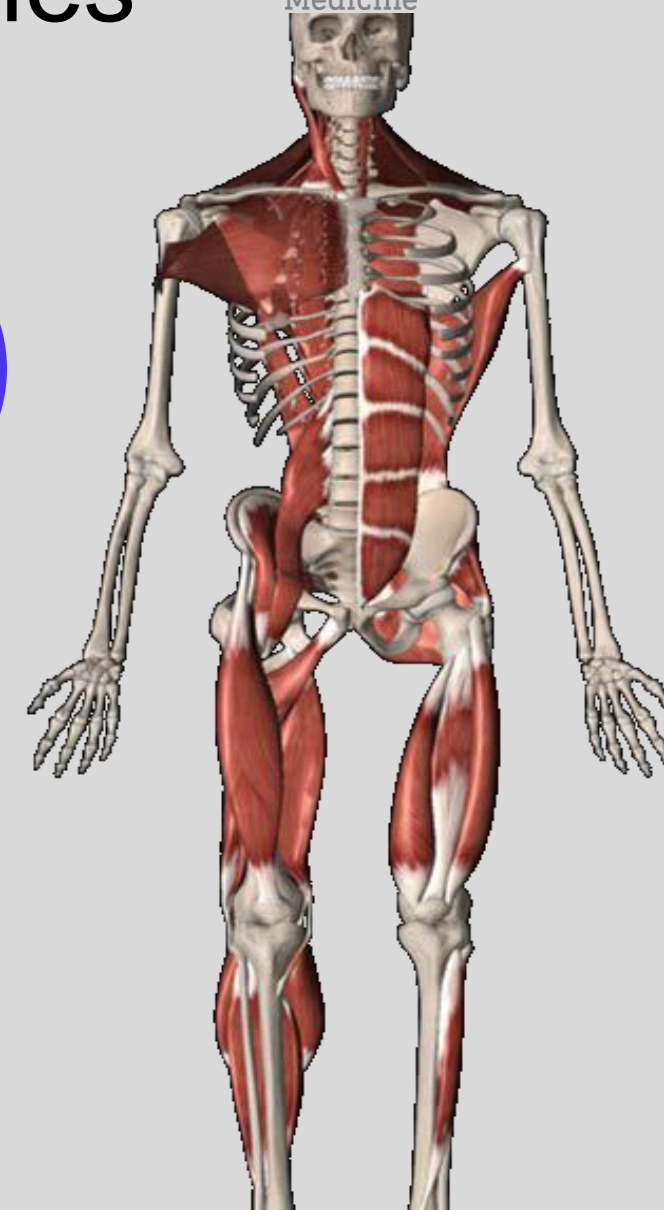


Table 2 – Posterior Muscles involved in Upper and Lower Crossed Syndromes

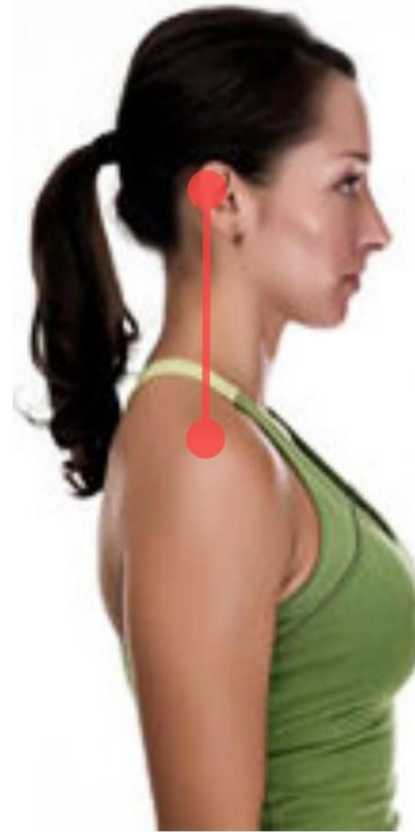
memorize

Posterior Postural Muscles	Posterior Phasic Muscles
Upper Cross	
Upper Trapezius	Mid-lower Trapezius
Levator Scapulae	Rhomboids
Cervical erector spinae	Latissimus dorsi
Lower Cross	
Quadratus lumborum	Gluteus minimus
Thoracolumbar erector Spinae	Gluteus medius
Hamstrings	Gluteus maximus
Piriformis	
Gastroc-soleus complex	



Clinical Association: HEAD FORWARD POSTURE and Upper Crossed Syndrome

- Increased thoracic kyphosis
Rounded shoulders
- Flexion of the lower cervical spine
- Extension of upper cervical spine
- Muscle imbalance –
tight/facilitated type I
muscles vs weak/inhibited
type II muscles



Correct Head Posture



Forward Head Posture

Clinical Correlation

IJSPT

CLINICAL SUGGESTION

CERVICOGENIC HEADACHES: AN EVIDENCE-LED APPROACH TO CLINICAL MANAGEMENT

Phil Page, PhD, PT, ATC, CSCS, FACSM

ABSTRACT

Cervicogenic headache (CGH), as the diagnosis suggests, refers to a headache of cervical origin. Historically, these types of headaches were difficult to diagnose and treat because their etiology and pathophysiology was not well-understood. Even today, management of a CGH remains challenging for sports rehabilitation specialists. The purpose of this clinical suggestion is to review the literature on CGH and develop an evidence-led approach to assessment and clinical management of CGH.

Key Words: Headache, neck pain, muscle imbalance

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The International Journal of Sports Physical Therapy | Volume 6, Number 3 | September 2011 | Page 254

Page, P. Cervicogenic Headaches: An Evidence-Led Approach to Clinical Management, *The International Journal of Sports Physical Therapy*, 2011 Sep, 6(3):254-266, PMID: [22034615](https://pubmed.ncbi.nlm.nih.gov/22034615/);

LINK: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3201065/>

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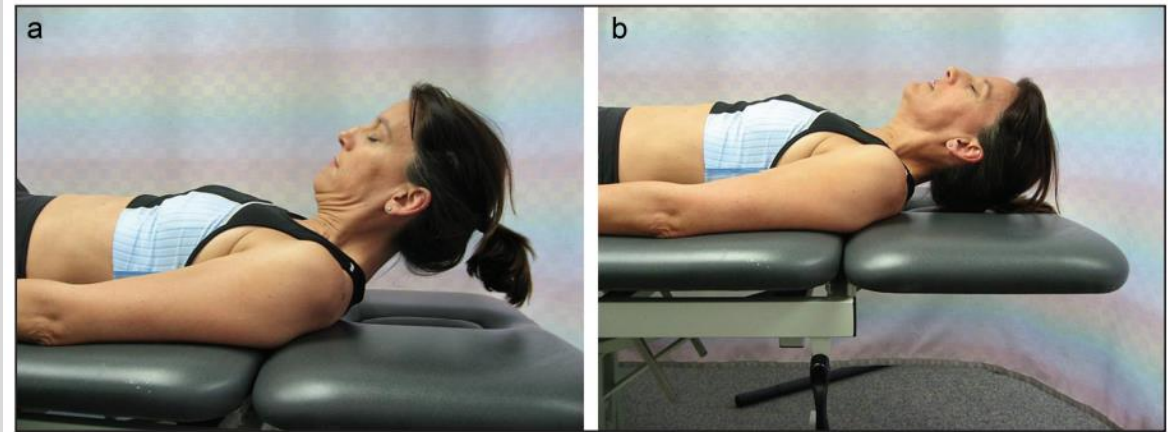


Figure 2a and 2b. Janda's Cervical Flexion test (a. Normal, b. Abnormal movement patterns)

The International Journal of Sports Physical Therapy | Volume 6, Number 3 | September 2011 | Page 257

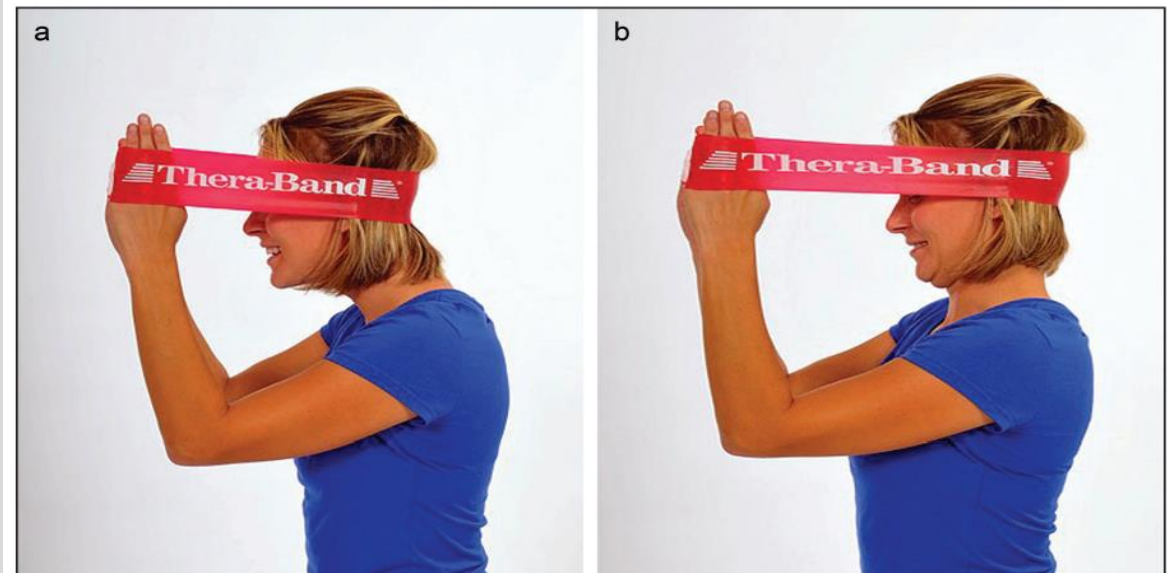


Figure 6a-b: Craniocervical flexion against elastic band loop. Begin with cervical spine in protraction (a). Maintain hand position while retracting against the tension in the band (b). (Used with permission of The Hygenic Corporation).

The International Journal of Sports Physical Therapy | Volume 6, Number 3 | September 2011 | Page 261

Other Postural Considerations

- **Postural Compensation as an “Integrated Homeostatic Process”:**

- Begins in utero, continues throughout growth.
- The CNS and peripheral nervous system together learn to balance the postural homeostatic mechanisms.
 - “Each person will progressively compensate in a different way depending, in part, on his or her unique intrinsic and extrinsic risk factors”
 - “Postural preferences and patterns will conform within the resultant connective tissue structure”
 - “Neuromusculoskeletal structures will remodel to reflect the functional demands placed upon them”
 - “Functional capability will be limited by less than ideal structure.”

Other Postural Considerations

- **Tensegrity Model:**

- From Architect Buckminster Fuller (1929) →
Tension + Integrity = Tensegrity; **this can also apply to biological systems**, in addition to architecture.
- “In this model, it is recognized that:”
 - “Areas of hypomobile somatic dysfunction may cause compensatory hypermobility elsewhere”
 - “Hypermobility in one or more regions may affect compensatory hypomobility elsewhere”
 - “Fascial patterning and musculoligamentous tensions take on primary importance as the model’s connecting elements”

Other Postural Considerations

- **Coronal Plane:** Groups Curves
- **Sagittal Plane:** Kyphoses and Lordoses
- **Transverse Plane:** Rotational preferences and Transitional Regions (OA, CT, TL, LS)
- **Correlate with:** Somatic Dysfunction

Other Postural Considerations

- **Gravitational Strain Pathophysiology (GSP)**

signs & symptoms:

- Changes/alterations in postural alignment
- Recurrent somatic dysfunctions
- Lower back pain
- Referred pain to extremities
- Headache
- Somatovisceral effects/presentations
- Fatigue
- Weakness

- Progression to: **Postural Decompensation**, “when an individual’s functional homeostatic mechanisms are overwhelmed or when the degree of pathological change becomes structurally incapable of resisting gravitational force.”

Seffinger, Michael A., D.O., executive editor, **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29**: “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera, And read from Heading of “Common Postural Clinical Conditions” (including the section on “Gravitational Strain Pathophysiology and Postural Decompensation”) to just before the Heading of “General Diagnostic Considerations”), LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547196>

Other Postural Considerations

- **Gravitational Strain Pathophysiology (GSP) Considerations:**
 - Intrinsic Factors
 - Extrinsic Factors
 - Tissue Responses:
 - **Skeletal-Arthrodial (Wolff Law: “bone remodels over time in response to the stresses placed upon it.” – (Foundations, 3rd Ed. p. 446)**
 - **Regional (Transition Zone) Fascial Response (J. Gordon Zink, D.O. and the Respiratory-Circulatory Model)**
 - **Muscular Response:**
 - TriggerPoints (may be related to myotomal or sclerotomal pain patterns)
 - Tight vs. Weak muscle patterns, such as **the Upper Crossed and Lower (Pelvic) Crossed Syndromes**
 - **Ligamentous Response**
 - **Response of Related Neural, Vascular, Lymphatic and Visceral Elements**

Other Postural Considerations

● Gravitational Strain Pathophysiology (GSP) Considerations:

from General Diagnostic Considerations:

- “The **purpose of postural diagnosis** therefore is to determine where in the continuum a given patient is and whether **the osteopathic postural model** could or should be integrated into that patient’s healthcare regimen. To this end, **the diagnostic process** should answer at least **four specific questions:**”
 - (1) “Does the patient’s posture play a role in a patient’s given complaint?”
 - (2) “Does the patient’s posture detract or limit the patient’s ability for self-healing?”
 - (3) “Does the patient’s posture present a significant risk factor for pain, dysfunction, or pathology in the future?”
 - (4) “Is the patient able to respond positively to a postural treatment regimen in a time- and cost-effective manner?”

Seffinger, Michael A., D.O., executive editor, Foundations of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, Fourth Edition, Wolters Kluwer, 2018, Chapter 29: “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera, And read from Heading of “Common Postural Clinical Conditions” (including the section on “Gravitational Strain Pathophysiology and Postural Decompensation”) to just before the Heading of “General Diagnostic Considerations”), LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547196>

Other Postural Considerations

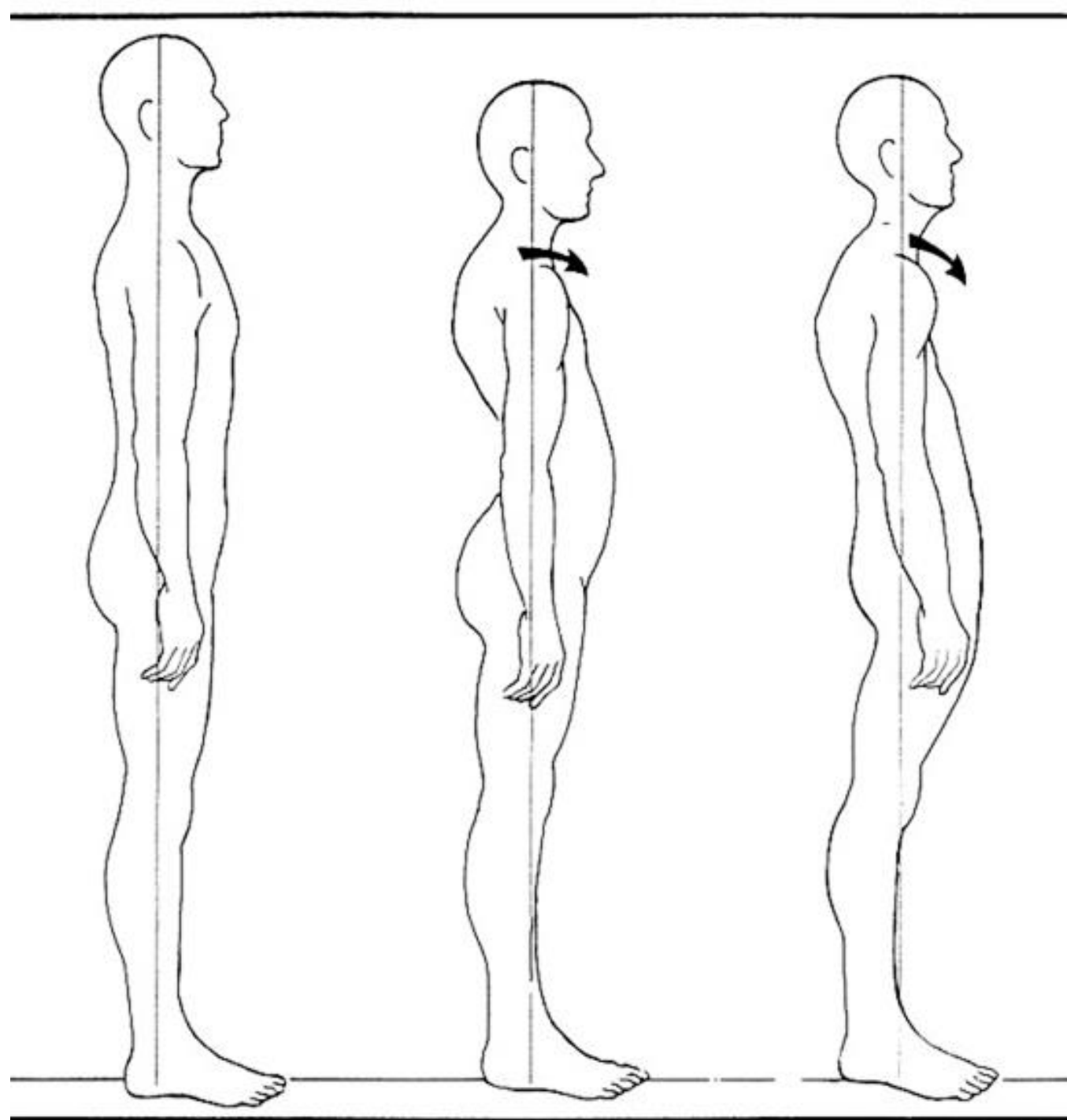
- **Gravitational Strain Pathophysiology (GSP)**
Considerations: from General Diagnostic Considerations:
 - “The **postural diagnostic approach** to answering these questions **relies heavily on observation and palpation**. It attempts to **correlate** other findings from the patient’s history and physical examination in order to determine if any pertinent signs or symptoms exist. It may also bring in **radiological or biomechanical tests** to help **quantify** the extent and type of postural change.”

Postural Considerations: General Treatment Strategies

- “(1) Postural Education and Compliance
- (2) OMT generally and to transition zones
- (3) Postural exercise
- (4) Static postural bracing
- (5) Functional postural orthotics
- (6) Other miscellaneous approaches”

- Symptoms-compensatory or facilitative mechanisms may cause:

- Cervical, thoracic, lumbosacral pain
- GI disorders, abdominal, pelvic and inguinal pain
- Headache, neck pain



General Treatment Strategies, cont.

Overall, there are many approaches that “are used to assist the body’s postural response to gravity. An **individually designed** conservative program includes **a carefully selected combination of patient education, OMT, exercise, and functional orthotics**. All of these are aimed at modifying the structure-function relationship and enhancing the body’s ability to self-heal. **Therefore, postural balancing requires an understanding of the biomechanical nature and functional anatomy of each patient and a full understanding of osteopathic philosophy.**

- “Patients are also awakening to the fact that current conventional medical interventions are often dangerous, costly, and ineffective.”
- “The U.S. medical system continues to be the third largest cause of death, after cardiovascular illness and cancer.”
- “To change this long-term trend, it is vital to employ therapeutics that are effective, efficient, and relatively less expensive and produce the least side effects.”
- “The profession needs to identify and treat the root cause of illness, be it of the body, mind, spirit, and/or the energy fields that support them.
- DOs need to develop, refine, and use their osteopathic skills so that they can adhere to Dr. Still’s reminder, ‘To find health should be the object of the doctor. Anyone can find disease.’”

Upper Crossed Syndrome: Exercises

Patricia Kooyman, DO, Michael Terzella, DO and
Joanne Donoghue, PhD

All of the following muscles play a role in upper crossed syndrome.

Upper
Crossed Syndrome
Stretching

1

Scalenes



Stabilize clavicle using your opposite side hand. Tuck chin, translate head posteriorly. Side bend the neck away from affected side. Rotate either way into scalenes. Hold 15-20 seconds.

2

Levator Scapulae



grab edge of bed or chair. stabilize scapulae of the side with hypertonic muscle. Induce flexion, side bending, and rotation away from the scapulae, stretching into the barrier.. Feel a stretch, but not pain. This can also be done while lying on your back. Hold 15-20 seconds. Repeat on other side.

3

Sternocleidomastoid Upper Trapezius



synergistic with upper trapezius. Stabilize clavicle and first rib using your opposite hand. Induce flexion, side bending away and rotation toward the affected side. Stretch into the barrier. This can also be done while lying on your back. Hold 15-20 seconds. Repeat on other side.

4

**Pectoralis major/minor
Doorway stretch**



Put hands at height of nose on
door frame
Lean forward
Hold 30-45 seconds, repeat
with hands 6 inches higher

5

**Pectoralis major/minor
Stomach/Chest Stretch**



Grab top or side of door frame
keep feet wide
Lean forward, pushing out
stomach
Hold 30-45 seconds

Upper
Crossed Syndrome
Strengthening

**Strengthening cervical flexors/
stretching cervical erector
18 spinae**



Lie on back on bed or floor
with little or no towel under
head

Make a double chin (by
tucking your neck) being
careful not to allow your
head to come off surface

Hold 5 seconds

Repeat 10 times

Deep cervical flexors
(chin tuck)



This can be done
against the wall and
with a tennis ball under
chin as well

**Mid-Trapezius,
Rhomboids – Wrists to
wall**

24



Lean back against wall
Put elbows at same height as
shoulders
Rotate arms back, trying to
touch wrists to wall
Hold 5 seconds, repeat 8-20
times

**Mid-Trapezius, Rhomboids
supine,**

25



thumbs up and lift up toward ceiling
The last few degrees targets the
rhomboids

Latissimus dorsi, Trapezius



- Stand with back against wall, heels on wall if you can
- Face palms forward and do “snow angels” reaching up and overhead as far as you can
- Try to keep arms on wall, but if they have to come forward, no big deal
- Do 5-10 times

Summary Slide

- The Upper Crossed Syndrome provides a clear framework from which to view a patient's postural compensations and subsequently target treatment approaches (*which can include an Exercise Prescription, as noted in the examples*).
- Other postural considerations are manifold and important to consider in a patient's clinical presentation.
- Application of the osteopathic perspective and the osteopathic philosophy can be beneficial for a patient's overall well being.

References

1. Seffinger, Michael A., D.O., executive editor, **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29**: “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera (*read under the Heading of “Muscular Response” and see Table 29.3 Examples of Postural (Hypertonic-Prone) Muscles and Symptoms, and Table 29.4 Examples of Phasic (Pseudoparesis-Prone) Muscles and Symptoms*, LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547285>
 - *And under the Heading of “Summary” see the “Clinical Pearls” Box Section as relates to Tables 29.3 and Table 29.4 as noted above, LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547738>*
 - *And read from Heading of “Common Postural Clinical Conditions” (especially including the section on “Gravitational Strain Pathophysiology and Postural Decompensation”) to just before the Heading of “General Diagnostic Considerations”), LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547196>*

References, continued

2. Heinking KP, Giusti RE, Hensel KL, et al., **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fifth Edition**, Wolters Kluwer, 2026.

- **Chapter 61:** “Dynamic Strain and the Bioenergetics Model” by JAN T. HENDRYX, **LINK:** <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456>
- **Chapter 92:** “Postural Considerations in Osteopathic Diagnosis and Treatment” by LISA A. DESTEFANO, MICHAEL L. KUCHERA, **LINK:** <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261515440&bookid=3456>

References, continued

3. Key, Josephine, A model of movement dysfunction provides a classification system guiding diagnosis and therapeutic care in spinal pain and related musculoskeletal syndromes: A paradigm shift – **Part 1**, Journal of Bodywork and Movement Therapies, 2008, 12, 7-21.
4. Key, Josephine, A model of movement dysfunction provides a classification system guiding diagnosis and therapeutic care in spinal pain and related musculoskeletal syndromes: A paradigm shift – **Part 2**, Journal of Bodywork and Movement Therapies, 2008, 12, 105-120.
5. Key, Josephine, The Pelvic Crossed Syndromes: A reflection of imbalanced function in the myofascial envelope; a further exploration of Janda's work, Journal of Bodywork and Movement Therapies, 2010, 14, 299-301.
6. Moore, Michele K., Upper Crossed Syndrome and Its Relationship to Cervicogenic Headache, Journal of Manipulative and Physiological Therapeutics, July/August 2004;27;414-420.
7. Morris, Craig E., Greenman, Philip E., et. al. Vladimir Janda, MD, DSc: Tribute to a Master of Rehabilitation, SPINE, 2006, Volume 31, Number 9, pp. 1060-1064.
8. Page, P. "Cervicogenic headaches: An evidence-led approach to clinical management," *The International Journal of Sports Physical Therapy*, 2011 Sep, 6(3):254-266, LINK: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3201065/>

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